Airbnb-Style Vacation Rental Marketplace — Architecture Diagram

Phase-wise production-scale architecture derived from the project README. The repository implements the frontend layer; backend, event streaming and infrastructure are proposed production architecture.

Phase 0 — Implemented Frontend Application

Phase 1 — Clients & Edge / Ingress

Phase 2 — Core Marketplace Microservices

Phase 3 — Events, Storage, Cache & Search

Primary Production Data Flow

Client	→	Cloudflare	→	API Gateway	→	Marketplace Services
Booking	→	Payment	→	Kafka	→	Availability / Reviews
Listings	→	PostgreSQL + Redis	→	Search / Discovery	→	OpenSearch
Media	→	S3	→	CloudFront CDN	→	Client

Architecture Notes & Engineering Context

Mapping the submitted frontend to the proposed production marketplace

Layer	Implemented in submission	Production-scale extension
Client/UI	React 19 + Vite 8, modular components, CSS design tokens, local state and normalized datasets.	React/Next.js clients for desktop, mobile/PWA and native apps.
Behavior	Scroll spy, booking card, calendar, guest stepper, Photo Tour and Lightbox.	Availability, booking orchestration, auth and payment services behind an API gateway.
Data	Local mock property, review and gallery datasets; no transactional backend.	PostgreSQL primary/replicas, Redis, OpenSearch and object storage.
Events	Client-side interaction feedback only.	Kafka event bus for booking/payment/listing events and asynchronous consumers.
Media	Local/static image assets for deterministic visual QA.	S3 + CloudFront with responsive WebP/AVIF delivery and asynchronous optimization.

Key Production Design Considerations

- **Double-booking prevention:** Redis distributed slot locking combined with optimistic concurrency/version checks in PostgreSQL.
- **Geospatial discovery:** OpenSearch geo-point indexing and bounding-box filters for map and nearby-stay queries.
- **Media pipeline:** asynchronous ingestion/optimization producing responsive WebP/AVIF assets delivered through CloudFront.
- **Scope boundary:** the submitted repository is frontend-only; the service and infrastructure layers are proposals, not implemented dependencies.

AI-Assisted Delivery Phases

Phase	Focus	Deliverable
Phase 0	Reconnaissance	Reference analysis, architecture plan, component hierarchy and risk analysis.
Phase 1	Core implementation	Design tokens, top-to-bottom listing composition and initial visual QA.
Phase 2	Photo Tour	Categorized full-viewport gallery, scroll locking and navigation.
Phase 3	Lightbox & accessibility	Keyboard navigation, focus trapping/restoration and overlay behavior.
Phase 4	Final validation	Micro-adjustments, interaction QA, build/lint verification and submission readiness.

Source: Project README — component architecture, technology stack, AI-assisted workflow, validation results and proposed production architecture.